

Om Sanjay Singhan

Brooklyn, NY | [LinkedIn](#) | 917-328-0100 | omsinghan25@gmail.com | [Portfolio](#) | [Github](#)

EDUCATION

New York University

New York, NY

MS in Computer Engineering (GPA: 3.89)

Graduation Date: May 2025

- Coursework: Data Structures & Algorithms, Artificial Intelligence, Advanced Machine Learning, Deep Learning, Big Data, Network Security, Computer Systems Architecture, Real-time Embedded Systems.

WORK EXPERIENCE

New York University

New York, NY

Research Assistant

September 2024 – Present

- Created a scalable visualization tool using Node.js and Socket.io, enabling 30% faster situational response and enhancing grid management efficiency.
- Achieved 50% improved data fetching efficiency by incorporating TimeScale to handle large datasets, supporting high fidelity modeling and real-time processing.
- Designed D3.js dashboards providing actionable insights into grid health and resilience, allowing 10+ metrics for Real time monitoring. Established continuous integration and deployment pipelines and explored predictive machine learning models, contributing to 15% improvement in outage prediction accuracy.

Aroris Health

Minneapolis, MN

Software Developer

May 2024 - August 2024

- Developed data encryption and validation scripts integrated into API routes to ensure HIPAA compliance, using Python and Firebase to securely handle and validate sensitive patient data across 1000+ records.
- Programmed and optimized 10+ backend and frontend features using Node.js, React, GraphQL, and TypeORM, improving data integrity and user experience.
- Troubleshooted and debugged 20+ critical bugs improving system stability and user experience by optimizing SQL queries and front-end components, leading to a 30% performance boost.

PROJECTS

Personal AI Fitness Tool Using Retrieval Augmented Generation (RAG)

September 2024 - November 2024

Python, Streamlit, Langflow, AstraDB, OpenAI

- Built a fitness application using Streamlit, integrating LangFlow workflows to generate personalized macronutrient recommendations and workout plans, achieving real-time, user-specific insights based on contextual inputs with a response time under 500 ms.
- Managed over 10,000 user data points using Astra DB, utilizing vector search for contextual retrieval, achieving a query accuracy of 95% for AI-generated responses.
- Utilized OpenAI models to process and analyze user queries, achieving a 98% accuracy rate in generating tailored responses, with an API integration latency of <300 ms.

Horizon - Banking Made Easy

May 2024 - August 2024

React, Next.js, TypeScript, Dwolla API, ShadCN, Appwrite, Plaid API, ChartJs

- Developed Horizon, a financial SaaS platform, integrating bank accounts via Plaid and enabling real-time transaction monitoring and secure fund transfers through Dwolla, improving transaction visibility by 40%.
- Implemented robust authentication and responsive UI using React Hook Form, Zod, and TailwindCSS, enhancing user interface performance and ensuring a seamless experience across devices.
- Streamlined transaction history management, including real-time updates, pagination, and filtering, boosting data retrieval speed by 25% and improving user engagement.

Enhanced Resume/CV Matching Using Doc2Vec and Deep Learning

January 2024 - May 2024

Python, Doc2Vec, NLTK, SpaCy

- Architected a job-matching system using Doc2Vec, achieving a 90% accuracy rate in semantic matching between resumes and job descriptions, streamlining recruitment for tech roles.
- Implemented advanced text processing techniques (tokenization, lemmatization) using NLTK and SpaCy, improving data handling efficiency by 25%, and integrated Gensim for a 20% boost in matching precision.

SKILLS & INTERESTS

Skills: Python, Java, JavaScript, TypeScript, HTML/CSS, SQL, Node.js, React.js, Next.js, Matlab, C++/C, Azure, TypeORM, GraphQL, MongoDB, AWS, Docker, Git, Jira.

Interests: Software Development, Full Stack, Machine Learning, Artificial Intelligence, LLMs, Cloud, Data Analytics, Data Engineering.